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Review article

Advances in functional metamaterials: bridging mechanical, acoustic innovations with multifunctionality and adaptive responses

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ABSTRACT

Functional metamaterials have emerged as a class of engineered materials exhibiting properties not found in nature, enabling unprecedented control over electromagnetic, acoustic, thermal, and mechanical wave propagation. Their multifunctional characteristics position them as critical enablers for aerospace, automotive, biomedical, and energy applications. However, despite rapid academic progress, practical adoption remains limited due to unresolved challenges. This review systematically synthesizes recent advances (2020–2025) in functional and adaptive metamaterials, emphasizing structural innovations such as functionally graded materials, honeycomb/cellular architectures, and bio-inspired designs. The analysis draws on comparative evaluation of reported studies, tabulated literature summaries, and classification of physical phenomena governing metamaterial responses. Key findings highlight progress in additive manufacturing for scalable fabrication, AI/ML-driven design for performance optimization, and the integration of multifunctional properties, including energy absorption, acoustic damping, and tunable stiffness. Case studies demonstrate applications in lightweight aerospace components, vibration control systems, biomedical scaffolds, and sustainable material solutions. A clear identification of research gaps, limitations, and practical constraints is provided, supported by comparative tables and schematic figures. While significant progress has been achieved, challenges persist in scalability, durability, sustainability, and data-driven design reliability. This review outlines practical and industrial implications, recommends hybrid manufacturing strategies, circular-material approaches, and collaborative frameworks, and emphasizes the transition of metamaterials from laboratory concepts to real-world engineering solutions.

Abbreviations

AI	Artificial Intelligence
AM	Additive Manufacturing
BBO	Black-Box Optimization
CVD	Chemical Vapor Deposition
DMA	Direct Memory Access
EV	Electric Vehicles
FGFRC	Functionally Graded Fiber-Reinforced Concrete
FGM	Functionally Graded Materials
FTIR	Fourier Transform Infrared Spectroscopy
ML	Machine Learning
MCU	Microcontroller Unit
RF	Radio Frequency
SEM	Scanning Electron Microscope
SPD	Severe Plastic Deformation
STL	Sound Transmission Loss
WAAM	Wire Arc Additive Manufacturing
XRD	X-ray Diffraction

1. Introduction

Metamaterials, engineered to exhibit properties absent in nature, have revolutionized diverse scientific and engineering domains by enabling unparalleled control over electromagnetic, acoustic, and thermal wave propagation [1–3]. The ability of the metamaterial concept had emerged from conventional designs aimed typically at electromagnetic wave manipulation. Following seminal works such as the study on Veselago's hypothesis of NRI materials, these efforts were made to lay down the foundational concepts for metamaterials. Functional metamaterials exploit structures to enable properties not found in constituent materials or in naturally occurring materials. They are one of

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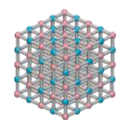
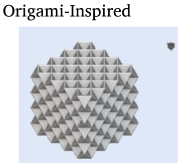
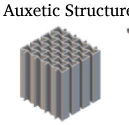
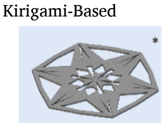
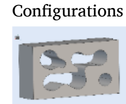
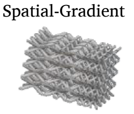
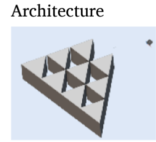
the leading advancements made in engineering technology that include dynamic features such as tunability and adaptability for the betterment of practical applications. Such advanced metamaterials use artificial intelligence, machine learning, and smart materials to further improve operational flexibility and performance. Examples range from the use of tuneable metamaterials, that can be controlled by external fields such as electric, magnetic, or thermal inputs, for tuneable lenses, cloaking devices, and vibration-damping systems, to the concept of active metamaterials comprising elements like varactors and phase change materials which have reshaped the control of wave behaviors under real-time conditions, such as beam steering and adaptive filtering [4–6]. Another significant advancement is the emergence of meta-devices, in which metamaterials are meticulously engineered into functional devices that facilitate energy harvesting, biomedical imaging, and wireless communication, thereby overcoming the limitations inherent in traditional designs. In addition, the integration of nanotechnology with metamaterials enabled the development of nanophotonic and nano-acoustic systems, where the manipulation of light and sound is performed at the nanoscale [7–9]. Such collaboration has led to revolutionary applications in superlenses, optical tweezers, and quantum information systems, marking the beginning of a new frontier in the further development of metamaterials. Table 1 shows the classification of a thorough foundation for understanding key geometry types, their adaptive features, and relevant potential applications across numerous fields.

Developments in additive manufacturing and 3D printing have propelled the field greatly, allowing for the construction of incredibly complex metamaterial structures with unparalleled resolution and scalability [10–12]. Programmable and digital metamaterials are presently being explored by researchers, wherein digital encoding and programmable logic are applied to realize real-time reconfigurability for superior wave manipulation. These breakthroughs illustrate the dynamic trajectory from traditional metamaterials to functional metamaterials, thereby highlighting their transformative potential in addressing challenges across the domains of science, technology, and industry.

Table 1
Classification framework of metamaterials.

Geometry Type	Structural Description	Adaptive Function	Application Field
Lattice-Based	Periodic unit cells	Energy absorption, Tunable stiffness	Automotive, Aerospace, sports Gear.
Cellular Topology 	Open/Closed foams, Trusses	Weight optimization, impact resistance	Biomedical, Packaging, Crash Safety
Origami-Inspired 	Foldable geometries	Reconfigurable, large deformation	Soft Robotics, Deployable Shelters
Auxetic Structures 	*Re*-entrant, Chiral	Negative Poisson's ratio	Protective Wear, Biomedical Scaffolds.
Kirigami-Based 	Cut-patterned Surfaces	Shape Morphing, Stretchability	Wearables, Stretchable Electronics
Hierarchical Geometry 	Multiscale Embedded Units	Multiband response	Sensors, Filters, Optics
Topological Configurations 	Edge-bound states	Defect-tolerant wave guidance	Acoustics, Phononic
Spatial-Gradient 	Gradient Cell distribution	Stress redirection programmability	Medical Implants, Smart Materials.
Fractal-Inspired Architecture 	Self-replicating geometry	Multi-frequency filtering	EM absorption, Heat Exchangers

1.1. *Integrating adaptive and multifunctional properties of metamaterials*

This is an age of innovation serving as an impetus for progress, wherein adaptive and multifunctional attributes in materials and systems have been seen as the bottom line for technological progress in virtually every domain. Adaptive characteristics define the responsiveness of materials or systems to changing conditions or stimuli or any operational requirements on the fly, whereas multi-characteristic attributes imply the ability to perform more than one role simultaneously, thus eliminating the need for many different special-purpose parts [13–15]. The convergence of these attributes effectively tackles the challenges related to efficiency, sustainability, and performance, and hence speaks to transformative potential across aerospace, healthcare, energy, and construction. Sustainable manufacturing processes have led to innovations in welding techniques that minimize waste and energy consumption [16,17]. Modern applications, especially in aerospace industries, demand materials that sustain extreme conditions such as temperature fluctuations or applied mechanical stresses without loss of structural integrity. For instance, materials for airplane wings or satellites can be designed to adjust their shape or stiffness to provide optimum aerodynamics or withstand adverse operating conditions, thus saving fuel and enhancing performance. Piezoelectric composites that have the capability of converting mechanical stress into electrical energy, thereby supporting energy harvesting along with a structural function, can contribute to realizing sustainability goals by minimizing waste and energy drawn from non-renewable sources [17–20]. The increasingly subtle sophistication of applications requires designing compact, light-weight, and multifunctional solutions; in electronics, for example, the need for miniaturization without sacrificing performance has pushed up innovation in terms of materials for integrated adaptive multifunctional properties. It can be applied to intelligent sensors, wherein self-healing property, electric conductivity, and sensing of environments are all accommodated in a single component to simplify device design while promoting functionality maximization. In wearable devices and implants, their accommodation of these functions has largely transformed patient monitoring and treatment in healthcare applications. Biosensors in wearable devices are equipped with adaptive sensing capability to respond to shifts in a patient's physiological states. Moreover, multifunctional coatings for implants enhance biocompatibility and inhibit microbial infection over long periods of time. Technologically, multifunctional properties' adaptation and integration are deemed a driving force behind innovation in robotics and artificial intelligence designs. Bioinspired soft robots combine adaptive sensing, like shape-memory alloy or hydrogels, that can replicate natural tissues' suppleness and responsiveness [21–23]. Besides mastering complex and delicate operations, these materials make robots resilient in unpredictable settings. These robots, particularly with their multifunctional properties, integrated sensory feedback mechanisms, and actuation systems-are viewed progressively as indispensable in the events of disaster response, surgery, and exploration.

Recent advancements in functional and adaptive metamaterials can be systematically categorized into five methodological classes: (i) material-oriented approaches, focusing on functionally graded composites, nanostructure reinforcements, and self-healing systems; (ii) structural or architectural approaches, including honeycomb, auxetic, chiral, and bio-inspired geometries for tailored mechanical and acoustic responses; (iii) manufacturing approaches, such as additive, hybrid, and nanoscale fabrication methods that expand design possibilities while addressing scalability challenges; (iv) computational and data-driven approaches, leveraging AI, machine learning, and topology optimization to accelerate discovery and predict complex behaviors; and (v) application-oriented approaches, which directly target aerospace, automotive, biomedical, and energy sectors. This classification provides a clearer state-of-the-art framework, within which the representative literature from 2020 to 2025 is summarized in Table 2.

Incorporating adaptive and multifunctional properties marks a novel development in energy within the sector [34,35]. Recent studies in functional metamaterials, spanning structural design, multi-functionality, and data-driven optimization, are summarized in Table 2 to highlight key developments from the last five years [1–5,12,22, 27–30]. For instance, smart coatings for solar panels have been engineered to change their reflective or absorptive rates according to variations in solar light intensity. On the other hand, multifunctional batteries and supercapacitors that combine the roles of structural support with energy storage represent another essential area in developing lightweight electric vehicles and solar-energy systems [36–38]. Such technologies meet needs for added functionality while at the same time conforming to the world's need to curb carbon emissions and mitigate climate warming. Benefits can be found for the construction and infrastructure industries. For instance, adaptive materials such as self-healing concretes can prolong structural lifespan by repairing stress cracks or water seepage, thus reducing costs of upkeep and enhancing safety factors. Their multifunctional properties, such as thermal insulation and vibration dampening, make them a center of focus in designing "smart" buildings that are efficient in their usage of energy. Their responsiveness to seisms, temperature variations, or moisture level changes aptly describes the current growing need for resilient infrastructure in view of negative climate change effects. Fig. 1 is an overview of adaptive mechanisms in metamaterials, showing stimulus-response relationships and multi-mechanism integrating strategies. External stimuli, including temperature, electric and magnetic fields, chemical concentration, and mechanical forces, activate distinct physical mechanisms such as shape memory, chemical swelling, and mechanical deformation. These mechanisms enable reversible and programmable material responses, such as shape change, strain, force generation, property switching, and deformation. On the right, multi-mechanism integration is illustrated, highlighting the potential for combining diverse adaptive strategies within a single architecture to achieve multifunctional and autonomous metamaterial behavior. Such a field has many of its own challenges for adaptive and multifunctional property developments [39–40]. Such activities demand interdisciplinary innovation fusing material science, engineering, and computational modeling to concoct novel, scalable, and cost-competitive solutions [41–43].

Table 2
Summary of recent literature (2020–2025) highlighting advancements in functional and adaptive metamaterials, including their structural innovations, multifunctional properties, and emerging applications.

Year	Author (s)	Focus Area	Key Contribution	Relevance to Current Work
2025	Gao et al. [24]	Crack-resistant metamaterials	Demonstrated damage-programmable metamaterials mimicking natural crack resistance	Highlights multifunctionality in structural integrity
2024	Bordiga et al. [25]	Nonlinear dynamic metamaterials	Automated discovery of reprogrammable nonlinear systems	Supports the adaptability theme
2025	Chang et al. [26]	AI-driven tunable metamaterials	ML-based design for on-demand noise attenuation	Shows AI/ML integration in wave control
2025	Ma et al. [27]	Graphene-based absorbers	Mid-IR tunable metamaterial absorber with high FOM	Demonstrates novel electromagnetic control
2024	Dalave et al. [28]	Meta-structures in AM	Energy absorption using AM-based meta-structures	Links to additive manufacturing
2025	Alkunte et al. [29]	Functionally Graded Metamaterials	Review of fabrication, modelling, and applications	Strongly connected to the FGM section
2024	Madenci et al. [30]	GFRP -reinforced FGMs	Improved bending and mechanical properties	Directly relevant to multifunctional composites
2025	Wei et al. [31]	Bioinspired soft robotics	Shape-memory alloy-actuated inchworm robot	Supports adaptive/ cognitive metamaterials
2025	Marciel et al. [32]	Flexible energy devices	Electrochromic energy storage with multifunctionality	Links multifunctionality with energy systems
2023	Mondal [33]	Chromism-induced energy storage	Novel smart storage materials with multifunctional roles	Extends the multifunctional energy absorption theme

State-of-the-art manufacturing techniques, such as 3D printing and additive manufacturing, would feature to mass-produce these with the necessary complexity and accuracy. However, for such to enjoy reliability and withstand real-life conditions, wide areas of study still exist. Additionally, understanding resultant long-term environmental implications related to recyclability and lifecycle usage of such materials becomes necessary for further usage of these same ones. This rising trend of adaptive and multifunctional properties represents a paradigm change in designing and deploying materials and systems. This approach places much emphasis on holistic solutions to address the increasing loads involved in contemporary society, while simultaneously addressing the limitations embodied in traditional approaches. This development widens the frontiers of innovation and calls upon it to achieve a balance between performance and sustainability. As researchers and engineers persist in their examination of the potential inherent in these advanced materials, the future presents a promise of smarter, more efficient, and environmentally responsible solutions that redefine the possibilities of technology and its impact on the world.

The aim of this review, after the introduction, is to look at and synthesize recent developments in the creation and application of functional metamaterials, specifically focusing on the development of metamaterials from traditional mechanical functions into multifunctional systems using acoustic and mechanical adaptive functionalities. By addressing the intersection of structural innovation and functional integration, the review aims to elucidate the transformative potential of metamaterials in solving complex engineering challenges and advancing technological frontiers. The major goal is to explain the different ways that rising design concepts and fabrication technologies have enabled these materials to exhibit unmatched flexibility, absorb forces, and respond to environments. The review is wide-ranging in its coverage, covering a wide variety of subjects that, together, serve to show the several possible functions of functional metamaterials. It considers advanced principles of design that control material construction of these types, such as functionally graded materials (FGMs), honeycomb structures, and other key architectures. These structural concepts are analyzed for their ability to balance structural strength and adaptive

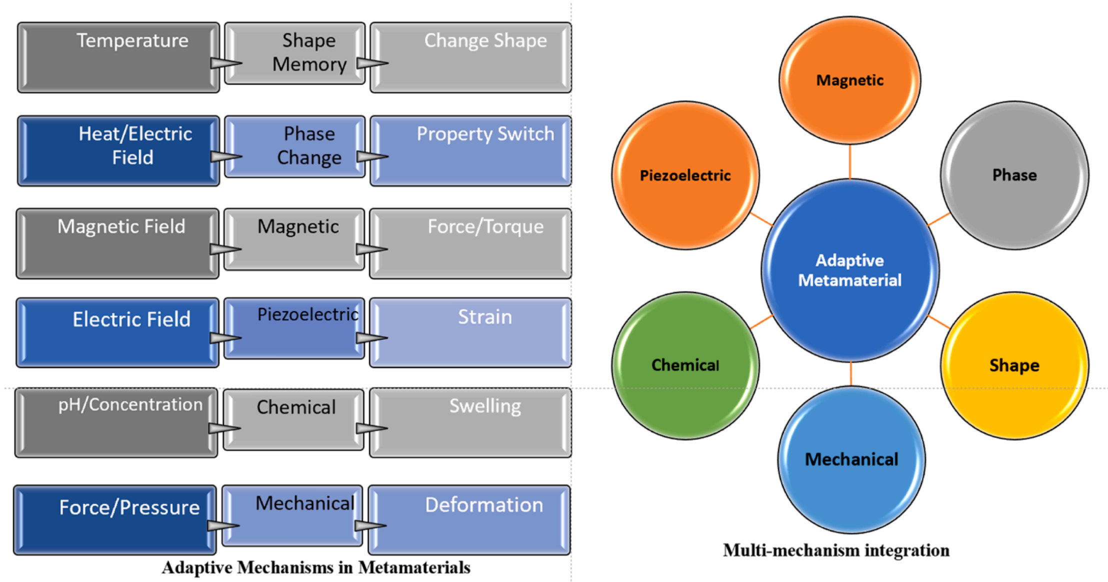


Fig. 1. Adaptive mechanisms in metamaterials, showing stimulus-response relationships and multi-mechanism integrating strategies. External stimuli activate physical mechanisms like shape memory, chemical swelling, and mechanical deformation, enabling programmable material responses, and on the right side, multi-mechanism integration demonstrates potential for combining diverse adaptive strategies within a single architecture to achieve multifunctional and autonomous metamaterial behavior.

ability and offer insights into how structural level modifications can lead to customized properties of performance.

In addition, the review explains that real-world applications of functional metamaterials can be seen in the development of core industries such as aerospace, medical, and automotive industries that utilize their unique characteristics, such as tunable stiffness, adaptive acoustic dampening, and multi-material layering to improve performance and sustainability. In such applications, the article highlights functional metamaterials' role in addressing emergent needs involving energy efficiency, noise reduction, material strength, and lifespan.

The focus of this work also includes the study of data-driven approaches like artificial intelligence (AI) aided modeling for the fine-tuning of design and optimization procedures of metamaterials. By combining computational technologies with traditional modes of design, researchers can anticipate and customize material properties with greater acuteness, thus shortening the developmental cycle and ensuring reliability of performance across different environments. Further, the review provides a future agenda direction by presenting emerging trends and demands of the field. It also highlights that future studies of this field must focus on developing metamaterials with cognitive and autonomous capabilities to enable their embedment in smart systems and sustainable technologies. The compilation reported in this review acts as a starting base for researchers, engineers, and policymakers to chart innovation and partnership directions in the fast-growing world of functional metamaterials. The next sections of this work discuss the current developments and their prospects. Section 2 explores innovations in material and structural technologies with special interest in functionally graded materials and their performance augmentations with honeycomb and cellular structures, known for their remarkable compressive and acoustic capabilities. In particular, it reviews acoustic and mechanical multifunctionality with great detail, with acoustic multifunctionality highlighting new acoustic dampening and insulating mechanisms, negative Poisson ratio-containing materials, and controllable stiffness. The use of adaptive and cognitive metamaterials is considered together with their sensory properties, as well as the significantly transformative potential of artificial intelligence and learning in engineering material design. Finally, the paper gives an overall application spectrum, emerging challenges, and concludes by pointing out the future directions in this field. An illustrative chart of the content is delineated in Fig. 2, including a brief overview.

This review has been written to bridge the gap between recent academic advances in functional metamaterials and their translation into real-world engineering applications. While several prior reviews have addressed specific aspects of metamaterials, there remains a need for a comprehensive synthesis that integrates structural innovations, data-driven design approaches, and practical industrial relevance. This article contributes by (i) consolidating recent literature from 2020 to 2025, (ii) linking adaptive and multifunctional properties with manufacturing and computational tools, and (iii) outlining practical implications, challenges, and recommendations. By doing so, it provides

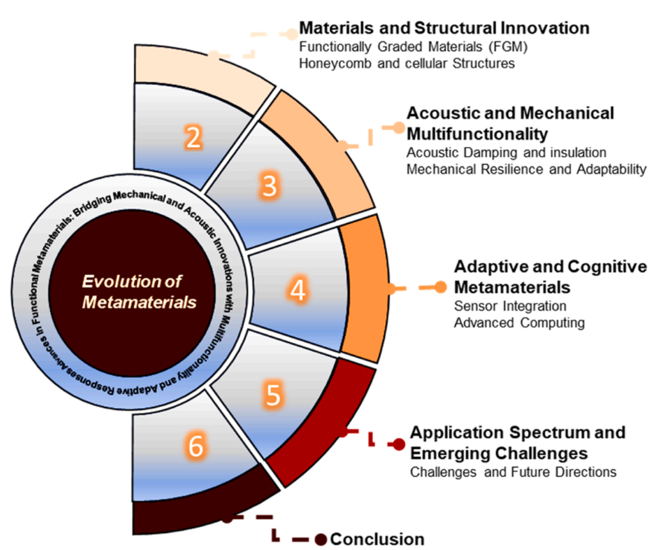


Fig. 2. Evolution of metamaterials: a schematic illustration showing the progressive development from (1) conceptual foundations, (2) electromagnetic metamaterials, (3) acoustic metamaterials, (4) thermal metamaterials, (5) multifunctional metamaterials, to (6) adaptive and reconfigurable metamaterials. The diagram emphasizes how the field has advanced by bridging mechanical and acoustic innovations with multifunctionality and adaptability, ultimately expanding its industrial relevance.

not only a foundation for researchers but also a practical reference for engineers and industry practitioners seeking to design and implement next-generation multifunctional materials. The motivation behind this review arises from the growing demand for multifunctional materials that combine structural, acoustic, and adaptive properties with manufacturability and sustainability. While prior reviews have addressed specific aspects of metamaterials, a holistic integration of structural innovations, physical phenomena, data-driven design, and practical implications is still lacking. The novelty of this work lies in (i) consolidating recent literature (2020–2025), (ii) highlighting physical mechanisms and research gaps, (iii) outlining limitations, challenges, and recommendations, and (iv) emphasizing industrial and sustainability-driven perspectives. By doing so, this review bridges theory and practice, serving as both an academic synthesis and a practical reference for engineers and industry professionals. The physical phenomena underpinning the behavior of functional metamaterials are diverse, ranging from resonance-based absorption and negative refractive index effects to tunable stiffness and nanoscale interactions. These phenomena determine the adaptability and multifunctionality of metamaterials across industrial applications. A consolidated summary of the major physical phenomena, their mechanisms, representative studies (2020–2025), and potential applications is presented in Table 3.

2. Material and structural innovations

The integration of hierarchical design, 3D printing, and multi-material architectures has pushed the boundaries of what metamaterials can achieve. By tailoring geometry at micro and macro scales, engineers can create systems with multi-functional capabilities, such as simultaneous acoustic insulation and mechanical strength.

2.1. Functionally graded materials (FGMs)

Functionally Graded Materials (FGM) are a family of advanced materials with changes in their composition, structure, and properties across their volume [58–60]. This customized gradient across the material allows for FGMs to mitigate many of the shortcomings typically found with conventional materials, such as stresses imposed by abrupt interfaces and problems of incompatibility. The status quo of FGM advancement reflects a confluence of innovative manufacturing technologies, pioneering applications, and spectacular increases in performance. First developed to accommodate the harsh thermal requirements of space applications, FGMs subsequently became multifunctional optimized materials for many industries, such as biomedical, transportation, and energy systems [61–64]. With respect to manufacturing, the evolution from common fabrication processes to next-generation additive manufacturing (AM) processes has transformed the processes of fabricating FGMs. While once processes such as powder metallurgy, chemical vapor deposition (CVD), and thermal spraying prevailed in this arena, these processes ensured enough control over the gradation of material, but ensured serious limitations with respect to complexity and tailoring. State-of-the-art AM processes such as Directed Energy Deposition (DED), Fused Deposition Modeling (FDM), and Selective Laser Sintering (SLS) now offer precise control of both material composition and shape [65–66]. These new processes allow for the smooth implementation of fine gradation in complex geometries, essentially extending boundaries of material engineering to new extremes. Advanced methods include laser-aided processes and gradient extrusion, which, in turn, have improved the capability of FGMs significantly by opening the way for localized customization of properties for specific functionalities. Applications of FGMs have diversified in parallel with advancements in their fabrication. In the biomedical sector, FGMs have become a cornerstone for implantable devices, particularly in orthopedics and dentistry, where their graded properties mimic the natural transition between hard and soft tissues [67–69]. Titanium-hydroxyapatite FGMs, for instance, have proven instrumental in addressing stress-shielding effects in bone implants [70]. Similarly, in a thermal barrier system, FGMs show excellent resistance to large temperature gradients, allowing for their use in gas turbines, rocket engines, and heat exchangers [71]. The automotive industry uses FGMs to decrease weight while improving crashworthiness and energy absorption for structural components [72–74]. Moreover, FGMs are critical in novel energy technologies, whose graded structures ensure efficient thermal and electrical conductivity, while being exemplified by solid oxide fuel cells and photovoltaic systems [75–77]. From a performance point of view, FGMs have established a benchmark in addressing the limitations inherent to conventional composite materials. Lack of distinct interfaces results in considerably lower interfacial delamination, and thus improves mechanical integrity. Graded properties also minimize thermal stresses, providing superior resistance to cracking and fatigue in high-stress environments. In addition, the ability to engineer specific property transitions, such as electrical conductivity or biocompatibility, has enabled FGMs to meet the stringent demands of modern applications. Current research in the area involves integration of nanoscale reinforcements, such as graphene and carbon nanotubes, in order to further improve their mechanical and functional properties [78–80]. Efforts to integrate machine learning and optimization algorithms into FGMs' design are also underway, thus able to make predictive modelling and accelerated development cycles possible. FGMs are at the forefront of material science innovation, evolving from a niche research topic to a critical enabler of high-performance applications. The synergy between advanced fabrication technologies, diversified applications, and tailored performance enhancements continues to drive their development. As industries continue demanding materials bearing multiple properties to overcome modern engineering challenges, FGMs represent a paradigm of material innovation, showcasing a continuous transition in properties that address the limitations of conventional composites while moving toward the multifunctionality and adaptability expected of the next generation smart materials. Fig. 3 highlights key aspects, including the state of FGMs (solid and liquid deposition), manufacturing methods (laser deposition, thermal plasma spray, centrifugal casting), and their intrinsic properties (chemical, physical, and mechanical). It also illustrates their compatibility and ability to inhibit crack propagation, leading to broad applications across biomedical implants, aerospace

Table 3
Summary of physical phenomena in metamaterials.

Phenomenon	Mechanism / Reaction	Reported Studies (2020–2025)	Applications
Resonance-based absorption	Localized resonators create selective frequency absorption	Chang et al. (2024) [4], Ma et al. (2024) [5]	Noise attenuation, IR absorbers
Negative refractive index	Simultaneous negative permittivity & permeability	Gao et al. (2024) [44], Bordiga et al. (2024) [45]	Cloaking, imaging
Phononic/ Photonic band gaps	Bragg scattering in periodic lattices	[46–48]	Acoustic filters, vibration isolation
Auxetic behavior	Negative Poisson's ratio deformation	[49–50]	Protective gear, biomedical scaffolds
Tunable stiffness	SMA, MR elastomers, LCEs reacting to stimuli	[51–56]	Robotics, aerospace, adaptive structures
Thermal-electromagnetic coupling	Phase-change or electrochemical tuning	Li et al. (2024) [6], Lo & Lai (2023) [57]	Thermal management, reconfigurable devices
Nano-acoustic/ photonic effects	Plasmonic & photo-elastic interactions at the nanoscale	Castillo et al. (2024) [7], Xia et al. (2024) [8]	Imaging, quantum systems

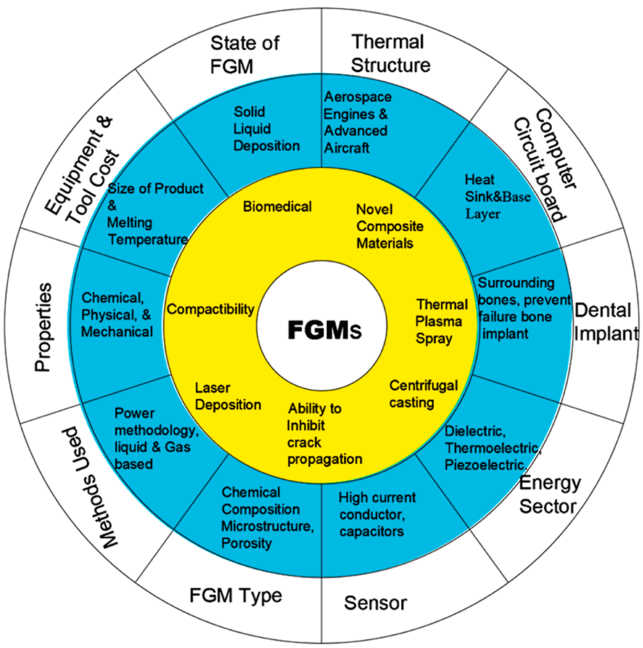


Fig. 3. Construction and application domains of functionally graded materials (FGMs).

structures, computer hardware (heat sinks, circuit boards), energy devices (capacitors, thermoelectric materials), and sensors. Table 4 illustrates the recent FGM research in terms of FGM techniques, application, and performance.

2.2. Design strategies for improved mechanical and acoustic properties: honeycomb and cellular structures

Honeycomb and cellular structures are widely used in modern engineering due to their exceptional mechanical efficiency and tunable acoustic properties. Fig. 4 depicts the honeycomb and cellular structures and their intricate architectures, inspired by natural systems such as beehives and biological cellular formations, which have been adapted and refined for use in aerospace, automotive, construction, and biomedical engineering [91,92]. The evolution of these structures from traditional hexagonal configurations to more complex geometries has been driven by advancements in computational modeling and additive manufacturing (AM), enabling the development of highly customized designs optimized for specific performance requirements. The state-of-the-art research in honeycomb and cellular structures emphasizes multifunctionality, where mechanical strength and acoustic performance are simultaneously optimized. Traditionally, honeycombs have been characterized by a hexagonal cell array and have received much research attention because of their high stiffness-to-weight ratios and lightweight properties. However, recent studies involved variations of cell geometry, re-entrant, auxetic, and graded designs for the enhancement of certain mechanical properties, such as energy absorption due to impact resistance and vibration damping. For example, auxetic honeycombs with negative Poisson's ratio characteristics are increasingly being used for applications requiring greater shear resistance and indentation strength [93]. Advances in material science likewise have played a critically facilitating role in developing cellular structures. Classical metal and polymeric honeycombs no longer stand alone but are reinforced with composite and functionally graded matter. These latter permit achievement of high stiffness, density, and acoustic impedance and, accordingly, offer constructions with unique performance specifications. For example, honeycombs reinforced with carbon fiber have demonstrated significant promise for applications requiring high strength-to-weight ratios, like aerospace panels and motor vehicles'

Table 4
Recent FGM research in terms of FGM techniques, application, and performance.

Manufacturing Techniques	Application	Performance	Reference
A simple casting method by incorporating 0–2 wt.% GPLs into epoxy resin, analyzed through SEM, FTIR, XRD, and DMA	The material is suited for biomedical tools, including in vivo stomach biopsy treatments and drug delivery systems.	Significant improvements in strength, stiffness, and thermal stability with a higher glass transition temperature	[81]
Severe plastic deformation (SPD) on powdered materials, involving material selection, mixing, cold compaction, SPD processing, and optional heat treatment, followed by characterization and testing.	High-strength engineering applications, offering adaptability across industries due to their gradual property transitions.	FGMs exhibit improved efficiency, durability, and performance, making them transformative in materials engineering.	[82]
FGMs are manufactured with gradient composition transitions between dissimilar materials, leveraging additive manufacturing (AM) and machine learning (ML) techniques for process optimization, defect detection, and real-time monitoring.	FGMs enhance component performance across industries by enabling location-dependent mechanical and physical properties tailored to specific requirements.	ML-based methods optimize FGM fabrication, improving accuracy, reducing defects, and enabling innovative designs with superior mechanical and physical properties.	[83]
Black-box optimization (BBO) was applied using the Optima framework, leveraging TPE, CMA-ES, and NSGA-II algorithms to optimize material composition for reducing residual thermal stress in multi-layered FGM plates.	The approach is relevant for designing FGMs with minimal thermal stress, aiding in complex engineering challenges in material science.	CMA-ES delivered superior optimization quality compared to TPE and NSGA-II, though TPE demonstrated faster convergence.	[84]
FGMs are fabricated using WAAM by controlling deposition parameters like wire feed rate and deposition speed, with post-processing techniques such as laser polishing, reducing surface roughness.	FGMs produced via WAAM are suitable for wear-sensitive applications, emphasizing enhanced functionality and reliability in large-scale, complex components.	Optimized wire feed rates improved tensile strength by 15 %, while post-processing reduced surface roughness by up to 80 %, ensuring superior mechanical and wear properties.	[85]
The thermal behavior of a semi-spherical porous FGM fin was analyzed using the Finite Difference Method (FDM) to solve nonlinear partial differential equations under periodic base temperature variations.	The study is relevant to optimizing fin designs in electronics cooling, HVAC systems, automotive engine cooling, and solar energy collection.	FGM II exhibited the most significant thermal distribution, with periodic heat transfer enhancing thermal efficiency through oscillatory base temperatures and grading parameters.	[86]
FGM samples were prepared using the hot compaction powder metallurgy	The materials are designed to improve ballistic performance	The N2 ($m = 2.5$) FGM sample showed 17.59 % improved ballistic efficiency	[87]

(continued on next page)

Table 4 (*continued*)

Manufacturing Techniques	Application	Performance	Reference
route, with a three-layered structure based on a power law index ($m = 1$ and 2.5).	against high-velocity ballistic impacts, specifically for use in military or protective armor.	and 13.35 % higher microhardness compared to the N1 ($m = 1$) FGM sample, with relative densities over 98 %.	
Forty-eight beam specimens (plain concrete, FRC, and FGFRC) were fabricated and tested for workability, compressive strength, split-tensile strength, and flexural behavior using a three-point bending test.	The study aims to evaluate the performance of Functionally Graded Fiber-Reinforced Concrete (FGFRC), potentially applicable in construction for more efficient and cost-effective fiber-reinforced concrete solutions.	The FGFRC beam with 0.9 % fibers exhibited similar flexural strength to the optimal beam (1.2 % fibers) but with 20 % less fracture energy. Gradation improved both flexural strength and fracture energy, showing superior performance compared to standard FRCs.	[88]
The paper introduces an adaptive phase-field fracture formulation with a cycle jump scheme to predict fatigue crack nucleation and growth, using Mori-Tanaka homogenization and an adaptive mesh refinement.	The approach is applied to predict fatigue crack growth in functionally graded materials under cyclic loading, offering insights into materials with high fatigue resistance.	The framework improves computational efficiency and accurately predicts fatigue crack growth, validated against experimental and numerical results with a focus on fatigue crack resistance across different material gradation directions.	[89]
The exploration utilized Wire Arc Additive Manufacturing (WAAM) combined with Gas Tungsten Arc Welding (GTAW) to create FGMs using ERNiCrMo-3 and ERCuSiA alloys.	The FGM is intended for applications requiring enhanced mechanical performance, with potential use in structural components benefiting from the combination of these materials.	The FGM exhibited a 120 % increase in microhardness, a 120 % increase in tensile strength, and a 60 % increase in yield strength, though corrosion resistance decreased with higher copper concentration.	[90]

crash absorbers. In parallel, functionally graded cellular matter is currently being studied for its potential to exhibit gradual property variations mechanically or acoustically throughout its construction to better mitigate stress concentrations and increase overall lifespan. The role of computational design and simulation tools has been transformative in optimizing honeycomb and cellular construction. Computational software like Finite Element Analysis and topology optimization enabled engineers to predict, before construction, such complex construction behaviors under loading and acoustics conditions [94]. With generative designing algorithms, these tools permit the design of complex cellular geometries that would not be possible with conventional manufacturing processes [95]. Parametric modeling, for example, has been used to design honeycomb geometries with graded density functions to yield superior vibration isolation and noise attenuation properties [96]. Computational strategies like these permit rapidity in arriving at designs while ensuring that these constructions exceed satisfactory performance and sustainability requirements. In consequence, additive manufacturing has played a determinant role in yielding access to advanced honeycomb and cellular construction merits.

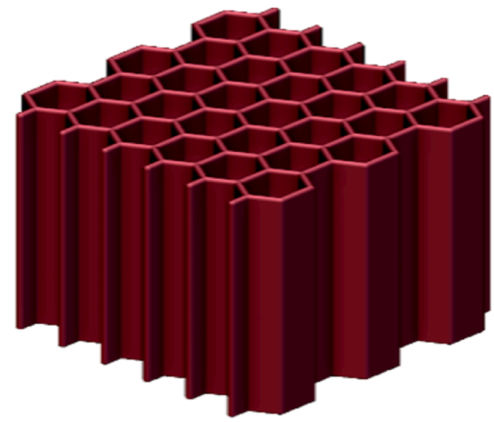


Fig. 4. honeycomb (hexagonal) lattice structure. This geometry is a representation of honeycomb metamaterials' lightweight but mechanically strong configuration, and they are heavily employed for purposes of absorbing energy, reinforcing structure, and multifunctional purposes, thanks to their high strength-to-weight ratio and efficient load transfer.

Methods like Fused Deposition Modeling (FDM), Selective Laser Sintering (SLS), and Direct Metal Laser Sintering (DMLS) allow for the manufacturing of complex geometries with little to no waste [97–101]. This has enabled expansion of the design space, such that hierarchical cellular architectures that combine macro-scale and micro-scale features can be developed with enhanced performance. For instance, lattice geometries with micro-architectural nodes have been shown to attain superior energy absorption capabilities and enhanced sound insulation capabilities [102]. In addition to that, 3D printing technology has made it possible to produce bioinspired cellular geometries, such as bone inner-architecture- or plant stem-structure-inspired ones, with remarkable strength and extensibility properties. The acoustics of honeycombs and cells has attracted extensive interest over recent decades, mainly for noise reduction and vibration control purposes. The ability to control wave acoustics through intentional designs of cells has led to designing acoustic wave-manipulative metamaterials with unprecedented sound attenuation behaviors [103]. Metamaterials with intelligent wave-manipulative capabilities mark a revolutionary leap in material engineering. Their adaptive, multifunctional, and programmable natures lead to innovations in almost every area of interest wherein wave involvement is integral [104]. Fig. 5 represents types of waves and mechanisms that intelligent metamaterials can administer and their real-life applications, and their engineering future.

The introduction of periodic or quasi-periodic arrangements within the cellular structure enabled the creation of photonic crystals and acoustic metamaterials [105]. These could be applied to block, redirect, or modify sound at specific frequency ranges on purpose. These innovations are finding applications in building acoustics, aerospace noise control, and even medical ultrasound devices. Recent work in the field also focuses on sustainability and the use of eco-friendly materials for making honeycombs and cellular structures [106,107]. The integration of biodegradable polymers, recycled materials, and bio-based composites into these structures aligns with global efforts to reduce the environmental impact of engineering solutions. Additionally, self-healing cellular materials that can restore their mechanical or acoustic properties after damage are a huge step toward long-lasting and more sustainable applications [108–110]. Challenges remain in fully exploiting the potential of honeycomb and cellular structures. For example, achieving a balance between lightweight design and structural integrity under dynamic loading conditions is an ongoing area of research. Similarly, the integration of multifunctional capabilities, such as mechanical strength, thermal insulation, and acoustic absorption in a single structure, remains to be further researched. Scaling up would be another

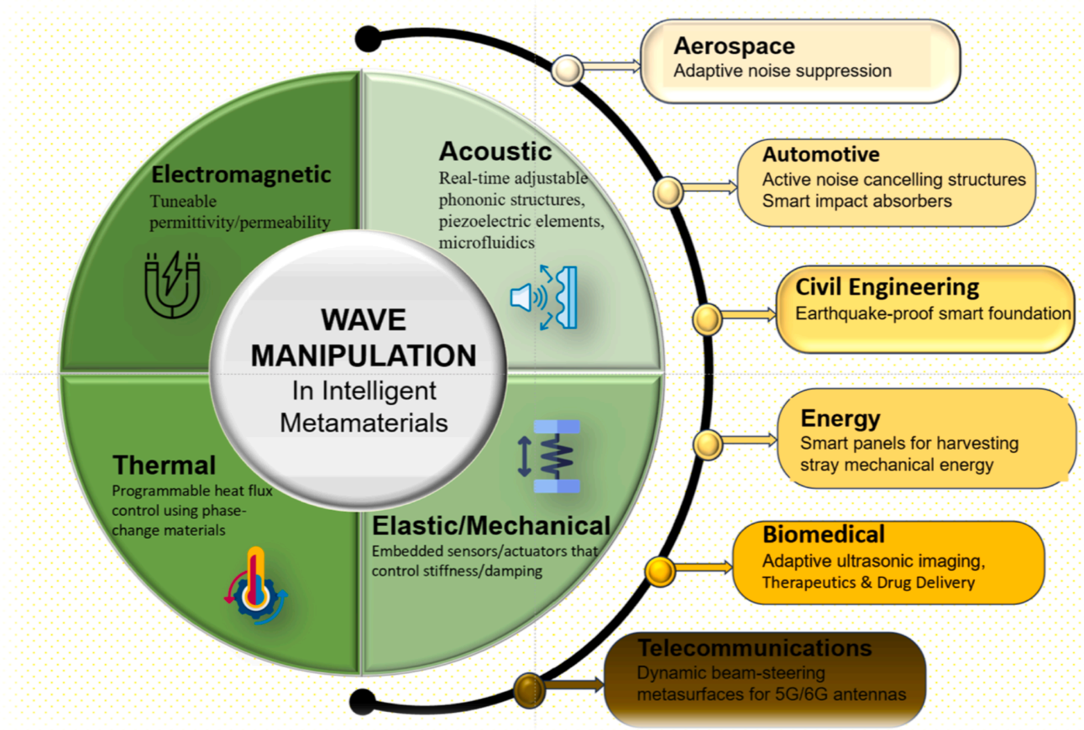


Fig. 5. Wave manipulation in intelligent metamaterials and their potential prospects in practical engineering. The diagram highlights different domains of wave control-electromagnetic, acoustic, thermal, and elastic/mechanical-enabled through mechanisms such as tunable permittivity, adjustable phononic structures, programmable heat flux, and embedded sensors/actuators. Potential applications span across multiple industries, including aerospace (adaptive noise suppression), automotive (active noise cancellation, smart absorbers), civil engineering (earthquake-proof foundations), energy (harvesting stray mechanical energy), biomedical (adaptive imaging, therapeutics, drug delivery), and telecommunications (dynamic beam-steering for 5G/6 G).

critical challenge; advanced cellular structures must be generated at a sufficient scale and cost-effectively while being precise in manufacturing. The future of honeycomb and cellular structures lies in their integration with emerging technologies such as artificial intelligence (AI) and machine learning (ML) [111,112]. Such technologies may be used to extend design parameters to improve considerably the ability to predict the cellular structure's performance and thus save development time as well as costs. Additionally, the incorporation of smart materials and sensors into cellular architectures holds promise for creating adaptive structures that can respond to changing environmental conditions [113]. Honeycomb panels made of piezoelectric materials are capable of modulating their stiffness or damping in real time, induced by an external stimulus, while unlocking new areas of applications in aerospace, robotics, health-related industries, etc. [114]. In conclusion, honeycomb and cellular structures represent a remarkable convergence of natural inspiration, advanced materials, and cutting-edge manufacturing techniques. Their evolution from traditional to multifunctional and adaptive architectures shows the abilities of these structures to solve the complex requirements of modern engineering applications. By continuing to explore innovative design strategies and leveraging the latest technological advancements, researchers and engineers can unlock new possibilities for improving the mechanical and acoustic performance of honeycomb and cellular structures, paving the way for their widespread adoption across diverse industries.

3. Acoustic and mechanical multifunctionality

Acoustic and mechanical multifunctionality is a paradigm where materials can perform multiple tasks across sound and mechanical domains. This approach allows designers to tailor properties like acoustic impedance, mechanical compliance, and energy dissipation, making them ideal for high-performance applications like lightweight aerospace structures and noise-cancelling enclosures [115–117].

3.1. Acoustic damping and insulation

Acoustic dampening and insulation are highly significant areas of material science and engineering, which focus on controlling and attenuating sound transmission in all environments [118]. There has been a growing quest for novel solutions due to rising urbanization, industrialization, and the need for quieter living and working spaces. All these developments require materials that are sustainable, efficient, and versatile across all applications, from the automotive and aerospace industries to single and combined residential and commercial building designs. Acoustic dampening is defined as the reduction in sound waves, thereby decreasing noise levels by converting vibration energy into heat. The opposite of this insulation is blocking or reflecting the transmission of sound, usually achieved through the creation of barriers that deny the ability of sound to pass through surfaces or structures [88]. Both mechanisms depend quite a lot on material properties such as density, porosity, elasticity, and design specifics of the structure to achieve the degree of noise control performance desired [119]. Modern research is focused on incorporating these principles into multifunctional materials capable of addressing a variety of acoustic and mechanical challenges. Multifunctional metamaterials that integrate acoustic dampening with mechanical resilience represent a giant stride in material science. By combining innovative structural designs with decoupled mechanisms, energy dissipation, and vibration control [120,121]. Fig. 6 summarizes key elements of other researchers, showing the overall layout emphasizes how acoustic and mechanical properties can be simultaneously engineered into a metamaterial system, representing a significant advancement over traditional materials that typically excel in only one functional domain.

Traditionally, the domain of fiberglass and mineral wool, for

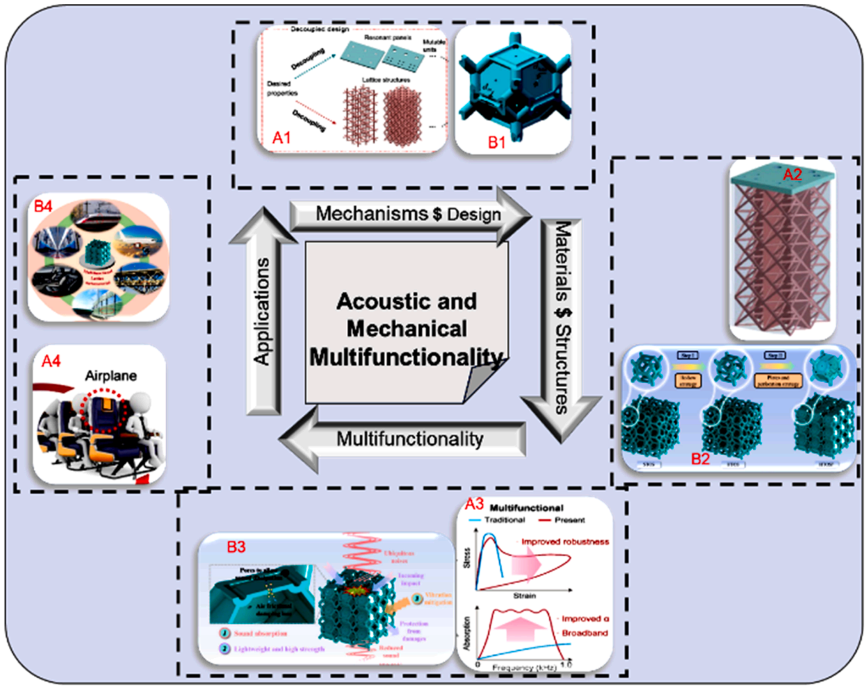


Fig. 6. Acoustic and mechanical multifunctionality in metamaterials. The schematic illustrates the interplay between mechanisms and design, materials and structures, multifunctionality, and applications. Representative examples are shown: (A1–A4) design strategies, multifunctional performance, and practical applications [121]; (B1–B4) structural architectures and material innovations contributing to multifunctional behavior [122]. Together, these examples demonstrate how metamaterials achieve enhanced robustness, broadband performance, and application-specific adaptability.

example, materials have dominated because of their inherent high porosity and sound absorption properties, making them ideal for use in acoustic insulation. Fiberglass and mineral wool work on the principle of conversion of sound energy into thermal energy by friction as well as by viscous effects resulting from the vibration of molecules of air within their dense yet porous structures [122]. In contrast, polyurethane and melamine foams offer lightweight and adjustable solutions with high absorption coefficients over a wide frequency range [123]. However, conventional materials such as these suffer from the drawbacks of being nonbiodegradable, problematic to individuals handling them, and restricted in their performance towards extreme conditions. Table 5 describes acoustic dampening and insulation in terms of mechanisms, materials, and innovations in recent research works.

3.2. Mechanical resilience and adaptability

Exceptionally resilient and adaptive materials have been the subject of transformative design developments in structures with negative Poisson's ratios, tunable stiffnesses, and responsiveness to external stimuli [133]. These innovative capabilities are gaining interest as engineering materials begin to create adaptable structures emulating those of biological systems while preserving accurate control over their mechanical behavior. The development of such materials is closely tied to their uses in aerospace, robotics, and biomedical devices, among other places where optimal performance is required under dynamic and unpredictable conditions. A recent area of material science concentrates on materials with a negative Poisson's ratio, also referred to as auxetics [134]. In contrast to other materials in which tension and strain lead to contraction at the transverse side, auxetic materials respond to tensile stress by being laterally expanded [135]. The paradox arises from specific microstructures that enable the coordination of deformation between directions. Mirroring natural systems that contain these auxetic structures in some specific biological tissues [136]. As such, there is inspiration to create synthetic auxetics through additive manufacturing and computational modeling. The new advances in auxetic materials have been toward tailoring their mechanical properties so that they can be applied within a specific application. Examples include protection gears and packaging, where auxetics greatly enhance impact resistance and energy absorption [137]. Finally, in biomedical engineering, they have promising potential for the design of dynamic stents and prosthetic structures that mimic adaptable anatomical structures [138]. Advances in techniques of fabrication attributable to 3D printing and nanoscale self-assemblies expand the space for design in auxetics significantly, allowing for complex patterns with tight control over Poisson's ratio [139].

Tuneable Stiffness is the fundamental ability that smart materials have a dynamically variation of stiffness, which allows for flexible behavior under different operative conditions [140]. Traditional materials, which possess fixed stiffness, are usually unable to meet the challenges created by modern applications as they require specific flexible responses to variations in loads, temperatures, or some other environmental inputs [141]. Materials capable of varying stiffness can fill the gap by changing their mechanical properties upon reaction to stimuli that can be temperature, electrical fields, magnetic fields, or mechanical loads. Of these, smart materials include shape memory alloys (SMAs), magnetorheological (MR) elastomers, and liquid crystal elastomers (LCEs) [142]. SMAs exhibit a stiffness with phase change behavior that is coupled to transformations triggered by temperature [143]. Similarly, MR elastomers exhibit a change in stiffness in a magnetic field, providing rapid, reversible tunability [144]. Advances in polymer chemistry and nanotechnology have led to the development of hybrid materials that include several mechanisms for stiffness modulations [145]. Such advances open possibilities both for applications in robotics, where the adaptive stiffness will improve mobility and safety, and in aerospace, where the materials endure extreme environmental fluctuations. Although significant advancements have been made, the field of mechanically adaptive materials is currently experiencing challenges in scalability, durability, and integration into existing systems. Also, although auxetic materials show promise, the field still awaits more development and manufacturing routes for ensuring

Table 5
Acoustic dampening and insulation: mechanisms, materials, and innovations.

Mechanism	Materials	Innovation	Ref
Active noise control uses electronic means to cancel sound waves, while passive noise control relies on materials and designs to absorb or reflect sound energy.	Sound insulation materials, including metamaterials, perforated panels, polymer composites, and recycled materials, are evaluated based on their sound transmission loss (STL) performance and practical applications.	Advancements in sound insulation technology feature predictive STL models, experimental methods, and eco-friendly solutions, driving future research and applications to mitigate noise pollution.	[123]
The acoustic performance of metamaterials is explained through properties like negative mass density, bulk modulus, and resonating microstructures.	Various acoustic metamaterials, including air-transparent options for ventilation, are highlighted to complement conventional sound insulation methods.	Combining frequency-specific meta-units with multi-layer conventional solutions offers enhanced sound insulation and addresses challenges like limited bandwidth.	[124]
Flat and uniform plate-like materials utilize the mass law for sound insulation, while acoustic metamaterials with periodic structures of resonators break the mass law at certain frequencies.	Traditional materials like rubber, glass, and metal require increased weight for better insulation, while polymer sheets integrated with spring-mass resonators enable advanced acoustic metamaterials.	High-throughput fabrication using polymer molding allows single-step implementation of metamaterial sheets, enabling customized designs for noise and vibration solutions.	[125]
The study investigates the role of acoustic and thermal insulation in enhancing indoor sound quality and reducing carbon emissions in green buildings.	Porous carbon sphere foam and high-quality building thermal insulation materials with low thermal conductivity are utilized.	The combination of structural configurations for construction boards and external walls, along with microporous foam particle systems, optimizes sound and thermal insulation performance.	[126]
Acoustic metamaterials (AMs) use periodic artificial structures to achieve extraordinary sound wave manipulation beyond traditional acoustic materials.	Membrane-type, plate-type, and smart-material-type sound insulation metamaterials, alongside composite structures and their interactions with boundaries and temperature effects, are highlighted.	Combining metamaterials with composite panel structures and optimization techniques offers lightweight, wide-band sound insulation solutions, advancing full-frequency performance applications.	[127]
The sound insulation mechanism of the double-layer metamaterial plate with multi-resonator (DLMPMR) is derived using the transfer matrix method and analyzed through finite element verification.	The proposed DLMPMR integrates lightweight and thin metamaterial structures with resonators to enhance sound insulation performance.	The DLMPMR achieves superior sound insulation by optimizing structural parameters to broaden resonance frequency ranges, shift resonance to lower frequencies, and exceed the mass law in the 220–5000 Hz range, demonstrating the potential for advanced noise control applications.	[128]
Gradient parameter multi-cell parallel synergetic coupling expands the working bandwidth for sound insulation.	A composite of transparent materials ensures sound insulation, vibration reduction, and light transmission.	Development of ultra-lightweight thin plate-type and high-stability thick plate-type metamaterials for varied applications.	[129]
The truss-based inertial amplification double-panel acoustic metamaterial forms a panel-cavity-panel vibroacoustic system that enhances sound transmission loss (STL) performance by incorporating truss structures with lumped masses and massless bars.	The metamaterial utilizes truss structures composed of two lumped masses and four rigid massless bars connected by moment-free hinges, alongside sandwich structures and an acoustic cavity.	A semi-analytical method based on the energy-generalized variational principle is developed for efficient STL prediction, and parametric studies are conducted to optimize sound insulation performance by adjusting the inertial amplification angle and cavity depth.	[130]
The study investigates the impact of spent graphite (SG) loading on the properties of rubber composites, focusing on their morphological, physical, thermal, mechanical, and acoustic behavior.	Two types of rubber (natural rubber and butyl rubber) were used as matrices, with spent graphite (SG) derived from lithium-ion battery waste as a sustainable reinforcing filler.	The incorporation of SG into the rubber composites, especially the BR/SG30 composite, led to significant improvements in properties such as density, hardness, thermal degradation, and sound insulation, offering potential applications in vibration and noise control.	[131]
The vibration-damping mechanism of the phononic crystal suspension (PCS) was studied through simulation and underwater tests, revealing the relationship between modes and attenuation zones (AZs).	Phononic crystals were used as the vibration-damping material to reduce the impact of internal actuating components' vibrations on sensors.	The PCS designed by Tianjin University demonstrated effective vibration-damping performance in an underwater environment, showing a maximum decrease of 6 dB at 2000 Hz.	[132]

scalable, quality-controlled processes. Similarly, materials with tunable stiffness provide a compromise between adaptability and long-term stability under cyclic loading conditions. Also, responsive structures have limitations in terms of energy efficiency due to resource-intensive actuation mechanisms. Emerging trends indicate that multi-material systems combined with design principles of hierarchy will greatly help overcome these challenges. Researchers from diverse disciplines intend to develop multifunctional systems emulating highly complex and efficient systems in nature by combining auxetic materials, adjustable stiffness, and responsiveness, all within a single framework [146–147]. Furthermore, advancement in sustainability and recycling will be called for so that the progress of adaptive materials is in line with ecological aims. In summary, the state of present developments in mechanical strength and adaptability presents a great deal of progress in negative Poisson's ratio materials, tunable stiffness technologies, and responsive structures. These innovations are going to tackle the existing engineering challenges as well as open new opportunities in various fields, such as health and space. Interdisciplinary collaboration and technological innovation will continue to be crucial for fully actualizing the potential of these new transformational materials.

4. Adaptive and cognitive metamaterials

Adaptive and cognitive metamaterials are advanced engineering materials that respond to external stimuli and adapt to their environment. These systems integrate sensing, computation, and actuation for autonomous decision-making. They are positioned to transform sectors such as soft robotics, wearable technology, aerospace, biomedicine, and telecommunications by integrating materials with machines.

4.1. Sensory integration

The infusion of artificial intelligence into material design fundamentally transformed how metamaterials with sensing and adaptive properties are developed [148]. It uses machine learning algorithms to scan vast datasets, searching for patterns and predicting the optimal configurations for specific performance metrics. An example of how AI-based generative design tools can quickly create an infinite number of iterations of material architecture to improve impact resistance, energy dissipation, and thermal conductivity. This data-driven approach significantly reduces process duration while still meeting strict performance requirements in the final stages of material design. In addition, artificial intelligence improves the predictability of the actual material behavior under complex loading and environmental conditions and thus allows for the ability to develop materials with autonomous, self-optimizing properties [149]. A major role in facilitating intelligent responses in metamaterials is played by innovation in structural engineering, in particular, through the employment of graded geometries and architected materials. Functionally graded designs can change from one material phase to another gradually; consequently, the gradients in density, stiffness, or porosity enhance energy absorption and stress distribution [150]. Other kinds of natural architectures, such as honeycomb and lattice structures, offer lightweight but mechanically robust solutions for energy dissipation and vibration control [151]. The integration of sensory elements, such as strain gauges or embedded fiber optics, within these systems enables real-time feedback whereby the material responds to environmental impacts dynamically. Incorporating sensory and adaptive traits in metamaterials opens up transformative applications across industries [152]. So, lightweight adaptive materials are used in aerospace for their high-impact strength, sound absorption, and thermal management in critical environments. In medical areas, intelligent materials with biocompatibility, having tunable stiffness, are applied in prosthetics, implants, and responsive drug delivery systems [153]. Automotive applications deploy adaptive acoustic damping and energy-absorbing structures to improve passenger safety while maintaining comfort, in addition to fuel efficiency [154]. In robotics, sensory-enabled metamaterials enable soft robotic systems to demonstrate higher dexterity and interaction with the environment [155]. Fig. 7 shows the important role of AI from different perspectives.

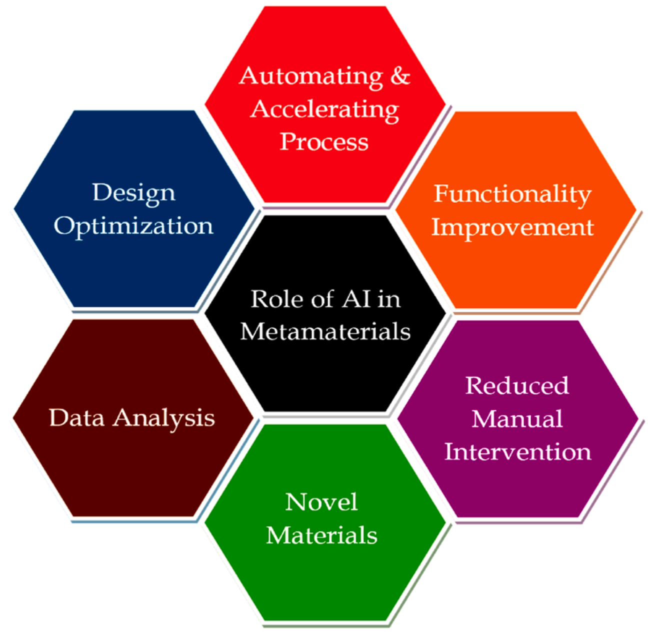


Fig. 7. Role of Artificial Intelligence (AI) in metamaterials. AI enables multiple advancements, including design optimization, data analysis, automation and accelerated processes, functionality improvement, reduced manual intervention, and the development of novel materials. These capabilities highlight AI's transformative potential in accelerating discovery, improving performance, and streamlining the deployment of intelligent metamaterials.

The development of responsive metamaterials to environmental changes becomes highly crucial as global industries are bound to shift toward sustainability. Researchers increasingly focus on incorporating bio-based polymers and recycling-friendly composites into materials [156–157]. For example, in addition to elongating the material's life, self-healing capabilities also minimize the need for resource-intensive repairs or replacements. In addition, lightweight and energy-efficient designs contribute to the reduced energy consumption of applications such as aviation and electric vehicles [158]. Those efforts align with overarching sustainability goals and excite environmentally conscious innovations in material science.

Notwithstanding the considerable advancements made, challenges persist in fully actualizing the potential of sensory and adaptive metamaterials. The upscaling of production while ensuring precision and cost-effectiveness presents a notable obstacle. Furthermore, the integration of multiple functionalities, without detriment to mechanical integrity, necessitates advanced design strategies and evaluations of material compatibility. Future efforts would most probably address developing self-sensing, adaptive, and communicating materials that will autonomously work with an external system, thereby acting as the foundation for implementing smart infrastructure and IoT-facilitated appliances and devices. The advancements in additive manufacturing and nanotechnology would also be seen as major contributors to creating highly sophisticated architectures with properties custom-designed for application. Finally, the integration of sensory and adaptivity properties into metamaterials establishes record-breaking levels of material science, engineering, and data-driven innovation. Advanced design paradigms are combined with composite material strategies and AI-driven optimization as researchers work on the development of intelligent materials that have transformative applications. This field has great potential impacts on strategically important industries in terms of performance, sustainability, and functionality, making it a foundational step toward a new era of smart, responsive, and sustainable technologies.

4.2. Advanced computing

The emergence of artificial intelligence and machine learning is revolutionizing the design and optimization of complex systems, especially in the domains of functional materials and structural engineering [159]. These capabilities critically empower the researcher to tackle the complexities of modern engineering, wherein competing requirements like mechanical adaptability, responsiveness to environmental conditions, and application-specific performance are difficult to simultaneously achieve through traditional methods. With AI and ML, researchers can now explore multi-dimensional parameter spaces with unprecedented precision, expediting the creation of novel solutions tailored to complex, real-world needs [160]. Indeed, previously, a design space for metamaterials was severely constrained by iterative physical prototyping and simulation; nowadays, paradigms for data-driven design create accelerations.

AI-powered algorithms, based on the model of generative adversarial networks and based on reinforcement learning, encourage the exploration of new geometries, which lets us build structures with adjustable mechanical and acoustic features. These models can predict metamaterial behavior under various operating conditions, optimized performance for energy absorption, acoustic dampening, or adaptive stiffness-and with minimized trial and error. One of the most influential and transformative areas of this kind is the application of artificial intelligence in multi-material composites to optimize these systems. With the use of machine-learning algorithms, researchers can find optimal material combinations and their spatial configurations to achieve highly customized mechanical properties [161]. For example, functional graded materials and honeycomb structures, which are known for their beautiful trade-off between lightweight design and structural integrity, can be optimized using machine learning-driven models. These algorithms optimize layer thickness, density distribution, and material compositions to meet stringent design criteria, ensuring mechanical robustness and environmental responsiveness simultaneously.

Another significant output of AI/ML is the acoustic capabilities of metamaterials, which progress far beyond previously fixed performance parameters of acoustic-damping materials in the past. AI-design models can thus provide excellent, tunable properties on adaptive acoustic metamaterials [162]. Backed by predictive analytics, such new technologies have particularly strong implications for industries such as aerospace and automotive, where control over NVH (noise, vibration, and harshness) happens to be of paramount importance. The integration of multi-physics solvers with optimization algorithms helps make the simulation workflows stronger so that acoustic manipulation capabilities do not degrade under varying operational conditions. Beyond performance optimization, AI and ML introduce cognitive and autonomous capabilities into material systems. The integration of embedded sensors and actuators with AI-driven analytics creates "smart" metamaterials capable of real-time adaptation to environmental changes. For example, in medical applications, AI-guided design can produce implants with self-adjusting stiffness to accommodate biomechanical loads, improving patient outcomes. Similarly, in sustainable energy systems, ML-driven designs can enable adaptive energy absorption in wind or tidal turbines, enhancing efficiency and longevity.

Another AI-enabled innovation is generative design. Generative design has transformed the way designers think about and prototype novel systems [163]. It uses algorithms to study the design space and develop a suite of viable solutions optimized for particular goals, such as weight minimization, mechanical strength, or cost-effectiveness. It best works in the aerospace and automotive industries, where the perennial challenge has been mass minimization in the absence of compromise on structural integrity. Accordingly, these designs are already transforming traditional supply chains into drivers of on-demand production of highly customized components in combination with additive manufacturing techniques. The interplay between AI, ML, and sustainable technologies offers challenges for reducing environmental footprints in new ways. For instance, one AI system could mine large datasets to identify eco-friendly substitutes for conventional materials and consequently streamline fabrication processes with less waste.

As an illustration, AI-driven LCAs could inform the strategic use of biodegradable polymers or recyclable composites in the production of metamaterials. Apart from this, ML algorithms can model the aging and degradation of the material, allowing for what amounts to proactive changes in the design toward longer product lives.

A cognitive metamaterial system architecture that represents an advanced intelligent sensing and control system that integrates artificial intelligence with adaptive metamaterial technology to create autonomous electromagnetic manipulation capabilities is illustrated in Fig. 8. This sophisticated system demonstrates how metamaterials can be enhanced with cognitive abilities for sensing, processing, and adaptive response to environmental changes [164]. The system initiates its operation through radio frequency (RF) wave capture via the sensing module [165]. The RF detector converts electromagnetic signals into digital data that flows to the processing unit [166]. The MCU with

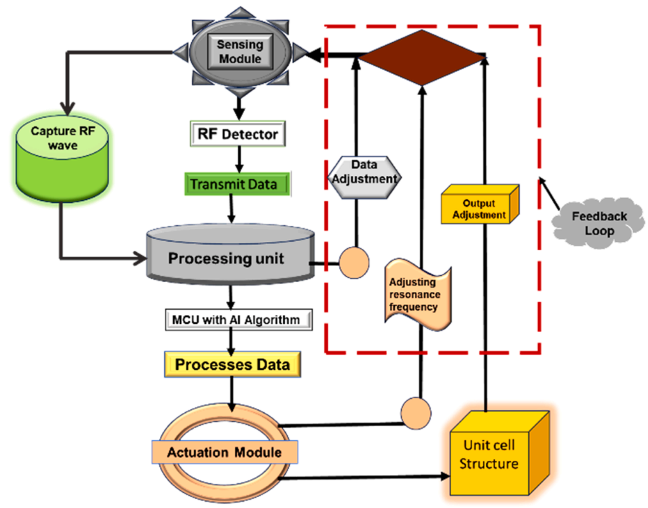


Fig. 8. Cognitive metamaterial system architecture. The schematic illustrates the integration of sensing, data processing, and actuation in an adaptive feedback loop. Radio-frequency (RF) waves are captured and detected, with data transmitted to a processing unit equipped with AI algorithms. The processed information drives the actuation module, which adjusts the unit cell structure by tuning the resonance frequency and output parameters. The closed-loop system enables self-adaptive, intelligent response of metamaterials for dynamic applications.

integrated AI algorithms processes this incoming data stream using trained machine learning models. Advanced signal techniques, including short-term Fourier analysis, enable the system to identify frequency components and environmental changes [167]. The AI algorithm within the processing unit analyses the sensor data to determine optimal system responses [168]. ML models enable the system to learn from past experiences and improve decision-making over time. The cognitive capabilities allow the system to predict environmental changes and proactively adjust its configuration. This intelligent processing distinguishes cognitive metamaterials from conventional passive metamaterial systems [169–170]. The system generates a control signal for the actuation module based on the AI analysis. The resonance frequency adjustment mechanism allows for the precise tuning of the metamaterial's electromagnetic properties [171]. The unit cell structure responds to actuation commands by physically reconfiguring its geometry or electrical properties. The reconfiguration process can involve mechanical deformation, electrical switching, or a change in material properties [172]. The feedback loop ensures continuous monitoring of performance and environmental conditions. The output adjustment mechanisms provide fine-tuning capabilities to optimize electromagnetic manipulation [173]. The system maintains a memory of successful configurations and can recall optimal settings for similar environmental conditions. This learning capability enables progressive improvement in the system performance over time [174]

5. Application spectrum and emerging challenges

This review explores advanced developments, emphasizing the profound changes they can bring about and future paths. Aerospace applications require materials that can adequately address both the reduction in weight and mechanical strength. Among various metamaterials, engineered with FGMs and honeycomb structures, such materials are highly important for achieving this balance. The development of adaptive acoustic dampening is also crucial in reducing noise produced by aircraft engines, thus enhancing passenger comfort and lowering environmental noise pollution [175]. Moreover, energy-absorbing structures, developed by the strategic layering of composites, have changed the paradigm of crashworthiness in aerospace design. By using

AI-enhanced modelling, researchers are now capable of simulating aerodynamic loads and thermal stresses to tailor the properties of specific aerospace components, such as fuselage panels and turbine blades. These developments help engineer self-healing and autonomous materials that dynamically react to in-flight stress, hence guaranteeing safety and efficiency in operations.

In the automotive industry, metamaterials serve to improve engine efficiency and the efficiency of vehicles. Tuneable stiffness mechanisms with multi-material layering enable components to adapt to changing load conditions to optimize energy use. The recent development of acoustic metamaterials has dramatically changed the design of interiors in vehicles, reducing cabin noise and enhancing sound insulation. At the same time, lightweight but strong materials are enabling safer and more energy-efficient electric vehicles (EVs). The use of AI-driven design tools pushes custom metamaterial architectures to the next level to ensure optimal performance levels in suspension systems, crash structures, and battery housings [176]. The integration of cognitive functionalities within metamaterials brings new opportunities for autonomous vehicles in future designs, where adaptive responses to environmental stimuli would significantly enhance both safety and user experience.

The paradigm under which the biomedical applications of metamaterials are changing is mainly driven by innovations that integrate mechanical robustness with biological functionality. Tuneable-mechanical-property metamaterials have shown great promise in advancing implants that imitate natural tissue behavior, such as bone scaffolds designed to promote osseointegration and minimize stress shielding. The abilities of acoustical manipulation enable the design of metamaterials for targeted drug delivery applications and sophisticated imaging techniques, further enhancing therapeutic possibilities. Functionally graded designs have also enabled the realization of prosthetic and orthotic devices with greater flexibility and comfort [177]. AI-enhanced modelling methods are now also playing a significant role in personalizing medical equipment to improve the performance of those medical devices for each patient. The future of biomedical metamaterials is based on the inclusion of bio-responsive and self-regenerative properties, which would advance the development of implants and devices that allow for autonomous healing and adaptation. Metamaterials are positioned at the forefront of enhancing environmental sustainability through innovative design and application. By using structural changes and composite integration, scientists can manufacture materials with perfect energy absorption and dissipation properties, thus leading to proper renewable energy systems. Using acoustic metamaterials is highly beneficial in controlling noise pollution, while the structure makes lightweight devices possible for environmentally friendly construction. The use of bio-based and recyclable materials in metamaterial fabrication holds potential solutions in the area of resource depletion and waste management [178]. AI-assisted approaches are enabling the efficient discovery of environmentally safe material compositions and their optimization for specific purposes. In the future, cognitive metamaterials capable of autonomous operation may play an essential role in ecological monitoring, offering real-time data about levels of air quality and water purity and self-adjusting to changing conditions. The potential of transforming metamaterials into practical functional materials is unarguable and fits across applications in aerospace, automotive, biomedical, and even environmental sustainability. These are materials that, through innovative design and incorporation of advanced technologies are changing the boundaries of what is possible in material science and engineering. While most researchers continue to probe the interplay of mechanical adaptability, energy efficiency, and environmental responsiveness, a metamaterial future would depend on its ability to integrate cognitive functionalities with sustainable practices. Once this gap between advanced engineering and practical applications is erased, functional metamaterials are expected to bridge opportunities toward a better future across different industries to secure a more sustainable and efficient future.

5.1. Knowledge gap

Despite the substantial progress achieved in the field of functional metamaterials, several critical knowledge gaps remain, which hinder their transition from laboratory concepts to large-scale industrial applications. A clearer articulation of these gaps is essential to guide future research and to emphasize the practical value of this review.

- Scalability of fabrication techniques: Although additive manufacturing, nanofabrication, and hybrid processing methods have enabled the realization of intricate metamaterial geometries, large-scale manufacturing remains costly, time-consuming, and restricted by limited material options [39,40]. Overcoming these scalability barriers is essential for widespread adoption.
- Insufficient long-term reliability data: Most reported studies focus on proof-of-concept demonstrations under controlled laboratory conditions. However, there is limited information on the long-term durability of metamaterials subjected to cyclic loading, thermal fluctuations, humidity, and corrosive service environments. This lack of reliability data restricts their direct application in aerospace, automotive, and biomedical industries.
- Integration of multifunctionality without compromise: The incorporation of sensing, actuation, or energy-harvesting functions into metamaterials often requires trade-offs in terms of structural integrity and mechanical robustness. The challenge lies in designing architectures that can achieve multifunctionality while maintaining strength, stability, and fatigue resistance.
- Data-driven design limitations: While artificial intelligence (AI) and machine learning (ML) have shown great promise in accelerating metamaterial discovery, their implementation is constrained by limited experimental datasets, high computational demands, and the challenge of model generalization across multiple length scales [41–42].
- Sustainability and recyclability concerns: Most high-performance metamaterials are still based on polymers, composites, or non-recyclable reinforcements, raising concerns about environmental impact and end-of-life scenarios. There is a clear need for research on biodegradable, recyclable, and self-healing metamaterials that align with circular economy principles [43].
- Limited translation to real-world case studies: Although numerous theoretical and experimental studies have demonstrated unique properties, only a few have progressed to practical prototypes in aerospace, automotive, energy, or biomedical sectors. Bridging this gap between academic research and industrial implementation remains a pressing challenge.

5.2. Limitations, challenges, and recommendations

5.2.1. Scalability and manufacturing limitations

Functional metamaterials have experienced significant advancements; however, their progression from laboratory prototypes to extensive applications poses considerable challenges [179]. A primary limitation resides in the scalability of manufacturing processes. The majority of metamaterials necessitate precision-engineered microstructures, rendering mass production both costly and time-consuming. Additive manufacturing has demonstrated potential in fabricating intricate geometries; however, its existing limitations in speed, resolution, and material diversity impede large-scale implementation [180]. Advances in scalable manufacturing techniques-including hybrid approaches that leverage additive and subtractive methods, will be a key component of future research focused on reducing the gap between designs with high performance and low cost. Therefore, new bonding techniques and materials that can maintain structural integrity while being able to adapt are needed. Nanostructure reinforcements and self-healing interfaces must become part of future research thrusts to advance these materials to higher lifespan capabilities in demanding applications.

5.2.2. Energy efficiency and power requirements

Energy efficiency is an important problem in the design of metamaterials, especially in the aerospace and automotive sectors [181]. Adaptive metamaterials, whether acoustic dampening or stiffness adjustment, often require external power sources to induce actuation. External power sources limit their integration into energy-efficient systems. Advances in energy harvesting and storage, such as piezoelectric and triboelectric, may provide a route toward developing self-powered metamaterials. Future research should focus on integrated energy systems, allowing metamaterials to operate autonomously and enhance their sustainability in green technologies.

5.2.3. Environmental sensitivity and durability

Environmental sensitivity, while an inherent property of smart metamaterials, brings along issues concerning durability and robustness. The time-dependent degradation in material behavior due to humidity, temperature, and chemical fluctuations can be avoided through bio-inspired designs mimicking nature's adaptive mechanisms. For example, water-sensitive polymers and temperature-sensitive composites have promise toward an environmental increase in adaptability, while their stability under long-term exposure remains questionable. Fig. 9 depicts the major key challenges of metamaterials.

Significant investigations into environmentally robust coatings and hybrid materials that remain functional across various conditions will be needed to expand applications in real-world environments.

5.2.4. Computational and data-driven challenges

Data-driven methods, including AI-enhanced modeling and optimization, revolutionized metamaterial design but also introduced challenges to computational scalability and accuracy. High-fidelity simulation of complex metamaterial structures is computer-intensive, especially when simulating large design spaces or multi-scale phenomena. Furthermore, the predictive accuracy of the artificial intelligence models relies crucially on good datasets, which are often limited for developing materials. Future research should focus on designing efficient algorithms and expanding the material database with experimental verification toward metamaterial design customized for different applications, it can be referred to in Fig. 10.

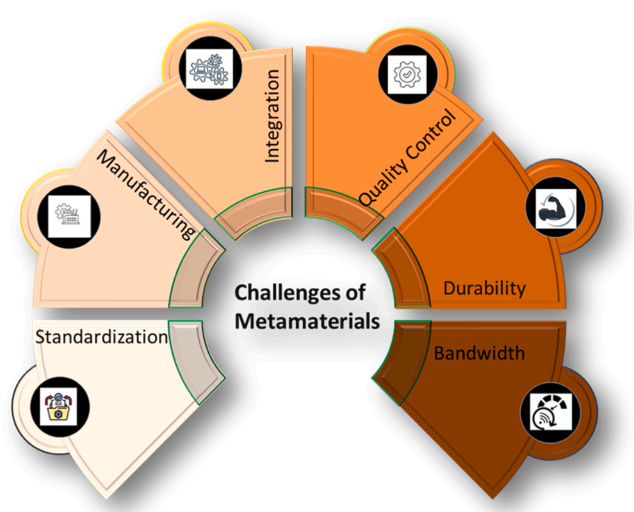


Fig. 9. Key challenges in the development and deployment of metamaterials. The schematic highlights critical barriers, including standardization, manufacturing scalability, system integration, quality control, durability, and bandwidth limitations. Addressing these challenges is essential for transitioning metamaterials from laboratory-scale demonstrations to reliable industrial and commercial applications.

5.2.5. Practical limitations and sustainability

Though there are immense developments, functional metamaterial research is confronted with some limitations. One such limitation is in the scalability of fabrication procedures. These conventional approaches, like additive manufacturing as well as nanofabrication, enable production for intricate geometries. However, they are offset by high expense, extended processing times, as well as a limited range of materials. Further, another limitation is the scarcity of durability information in realistic service conditions for cognitive as well as adaptive metamaterials subjected to cyclic loading, variations in temperature, or corrosive service environments. There are also issues in implementing multifunctionality without compromising structural integrity. Additions of sensing, actuating, or energy-harvesting functionalities often require a compromise between mechanical performance as well as tunability. Data-driven design methodology approaches, in principle great approaches have issues with low available datasets, computational cost, as well as model generalization at multiple length scales. Sustainability is an issue since most high-performance metamaterials are polymer-, composite-, or non-recyclable reinforcement-based materials whose end-of-life scenarios are not certain.

To tackle such problems, a number of recommendations are made. Firstly, it is necessary to develop hybrid manufacturing strategies that combine additive, subtractive, and bio-inspired technologies in a bid to add scalability while ensuring accuracy. Secondly, experimental verification over the long term in combination with accelerated durability testing is necessary for practical implementation in aerospace, biomedical, and automotive fields. Thirdly, machine learning and artificial intelligence integration needs to come in combination with open-access dataset creation for increased model credibility and reproducibility. Fourthly, a strong focus needs to be made towards green and circular-material design by adding biodegradable polymers, recyclable composites, and self-healing chemistry in a bid to ensure environmental compliance. Lastly, collaborative research between material scientists, mechanical engineers, computational designers, and sustainability professionals is bound to contribute towards translating lab discoveries into practical realities.

Fig. 11 highlights critical barriers, including scalability, durability, energy efficiency, environmental sensitivity, data/AI limitations, and sustainability (end-of-life). These factors collectively constrain the large-scale adoption of functional and adaptive metamaterials and underscore the need for hybrid manufacturing, eco-friendly materials, and long-term durability testing. Recent studies have demonstrated encouraging progress toward addressing these sustainability challenges. For example, PLA-based composites reinforced with carbon nanotubes have been explored as biodegradable and recyclable scaffolds for biomedical applications, combining mechanical robustness with eco-friendly degradation pathways [48]. Graphene-enhanced thermoplastic polymers have also been reported as recyclable materials for lightweight structural components, offering a balance between high strength and circular use [30]. In addition, self-healing elastomeric metamaterials have been developed to extend service life by autonomously repairing damage, thereby reducing material waste [111]. Emerging work on closed-loop recycling of thermoplastic composites and bio-inspired manufacturing approaches further illustrates practical routes for integrating recyclability into functional metamaterials. These case studies demonstrate how sustainable material strategies can be realistically incorporated into advanced metamaterial systems.

5.3. Practical and industrial implications

The advancements in functional and adaptive metamaterials carry wide-ranging implications for practical engineering and industrial applications. In the aerospace sector, lightweight yet resilient metamaterials with tunable stiffness and energy-absorbing capabilities can enhance fuel efficiency, structural safety, and noise reduction. For instance, functionally graded composites and honeycomb structures are

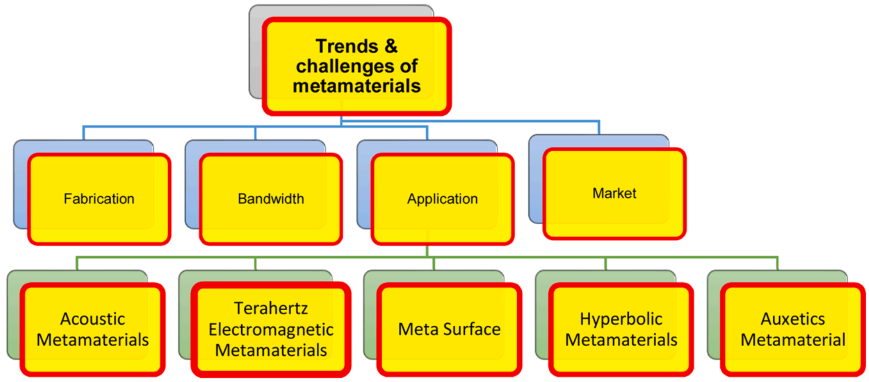


Fig. 10. Future directions and challenges of metamaterials. The schematic outlines key research and industrial challenges, including fabrication scalability, bandwidth limitations, application integration, and market adoption-alongside emerging metamaterial classes such as acoustic, terahertz electromagnetic, metasurfaces, hyperbolic, and auxetic metamaterials. Addressing these factors will be critical for advancing metamaterials toward practical, large-scale implementation.

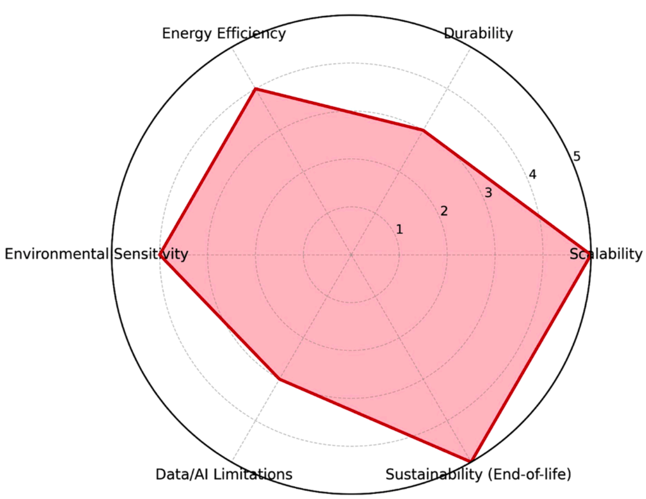


Fig. 11. Practical limitations and sustainability challenges of metamaterials.

increasingly being investigated for aircraft panels, turbine blades, and vibration-damping components. In the automotive industry, acoustic metamaterials enable improved cabin noise control, while multifunctional crash-absorbing meta-structures improve passenger safety in next-generation electric vehicles. Moreover, cognitive metamaterials with embedded sensing functions can support the development of autonomous vehicle systems with real-time adaptability.

In the biomedical field, metamaterials with tunable mechanical and acoustic properties provide opportunities for implants, prosthetics, and tissue scaffolds that mimic natural biological structures. These innovations can reduce stress shielding in bone implants, enhance osseointegration, and enable responsive drug delivery. The energy sector also benefits from metamaterials with multifunctional roles. Adaptive coatings for solar panels, structural batteries, and multifunctional supercapacitors integrate mechanical strength with energy storage, thereby supporting the transition toward lightweight renewable energy systems. From an industrial perspective, the integration of artificial intelligence (AI) and machine learning (ML) in metamaterial design is enabling data-driven optimization, reducing design cycles, and lowering the costs of prototyping. Additive manufacturing further accelerates scalability by enabling the rapid production of complex structures with minimal waste. These trends indicate that metamaterials are poised to drive disruptive innovations in sustainable manufacturing, healthcare technologies, and energy-efficient infrastructure.

The growing versatility of functional metamaterials enables their deployment across diverse industrial domains. In the aerospace sector, lightweight adaptive panels, vibration-damping fuselage structures, and advanced thermal management systems have been proposed to reduce fuel consumption and enhance safety under extreme operating conditions. The automotive industry has adopted metamaterial-inspired crash absorbers, acoustic silencers, and tunable suspension systems that combine energy efficiency with passenger comfort. In biomedical applications, biodegradable scaffolds with controlled porosity, implants with tunable stiffness, and smart drug-delivery platforms are among the most promising areas where metamaterials offer solutions to long-standing clinical challenges. Within the energy sector, metamaterials are being investigated for thermal barrier coatings in turbines, vibration control in wind and wave energy devices, and self-powered harvesting systems based on piezoelectric or triboelectric mechanisms. In civil engineering and architecture, seismic protection layers, noise-reducing wall panels, and adaptive building skins inspired by bio-mimetic principles represent emerging applications with direct societal impact. By linking the recent literature and classifications to these practical domains, this review underscores the transformative role of metamaterials in bridging fundamental research with industrial solutions. The rapid expansion of applications also reinforces the need for scalable fabrication, long-term durability, and sustainable material choices, which remain critical directions for future research.

6. Conclusion

The functional metamaterials domain has reached a critical juncture, which combines conventional mechanical properties with adaptive and multifunctional capabilities. This paper sheds light on highly important developments in this area, including functionally graded materials, honeycomb structures, and multi-material layering techniques, to offer tailored mechanical robustness and responsiveness to environmental changes. R&D in these areas has led to innovations that unlock tremendous opportunities across industries, be it aerospace, medical, or automotive. The introduction of artificial intelligence into data-driven design methodologies has radically changed material innovation, enabling very precise solutions for complex challenges. Such transformations open avenues for high-performance materials with sustainable applications that perform in multifaceted ways, responding to contemporary demands with unprecedented adaptability.

- Functionally graded materials and multi-material designs serve to enhance mechanical performance and adaptability.
- Advanced functions such as stiffness tunability and energy absorption enhance the application scope.
- AI-design methods liberate material precision and optimize.
- Cognitive metamaterials, integrating sensing and actuation, promise transformative impacts.

- The future of metamaterials is based on smart, sustainable, and scalable inventions.

CRediT authorship contribution statement

George Boafo: Writing – original draft, Methodology, Investigation, Conceptualization. **Deepak Kumar Biswal:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

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Data availability

Data will be made available on request.

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